Section 17: Dynamic Routing Protocols

Layer: L3 | Reference: Odom Vol1 Ch21: OSPF Concepts (protocol comparison context) | *Original study notes — CCNA 200-301*

Dynamic routing protocols let routers automatically learn and adapt to network topology changes instead of relying on manually configured static routes. The CCNA exam expects you to compare protocol families and understand convergence and route-selection concepts that apply across all of them.

Key Concepts

- Distance Vector protocols (e.g., RIP) share their own routing table with neighbors and trust what neighbors report without a full topology view
- Link-State protocols (e.g., OSPF) build a complete map of the network (LSDB) and independently calculate best paths using Dijkstra's algorithm
- Advanced Distance Vector / Hybrid (EIGRP) combines distance-vector simplicity with faster convergence using the DUAL algorithm and partial updates
- Path Vector protocols (BGP) make routing decisions based on path attributes and AS-path information, used between autonomous systems (the only EGP in common use)
- IGP (Interior Gateway Protocol) operates within a single autonomous system (OSPF, EIGRP, RIP); EGP (Exterior Gateway Protocol) operates between autonomous systems (BGP)
- Administrative Distance ranks trust in a routing source when multiple protocols learn the same destination: Connected=0, Static=1, eBGP=20, EIGRP=90, OSPF=110, RIP=120, iBGP=200
- Metric is protocol-specific and used to choose the best path WITHIN a single protocol: RIP uses hop count, OSPF uses cost (bandwidth-derived), EIGRP uses a composite of bandwidth and delay by default
- Convergence is the time it takes all routers to agree on the new topology after a change; link-state and advanced distance-vector protocols converge much faster than classic distance-vector
- Classful protocols (RIPv1) don't carry subnet mask info in updates; classless protocols (RIPv2, OSPF, EIGRP) do, enabling VLSM and discontiguous networks
- Split horizon prevents a router from advertising a route back out the same interface it was learned on, reducing loop risk in distance-vector protocols
- Passive interfaces suppress routing protocol hellos/updates on an interface while still advertising its connected network - common on LAN-facing interfaces

Command Reference

Command	Purpose
show ip route	Display the routing table, including the source/protocol code for each route (O=OSPF, S=static, C=connected, etc.)
show ip protocols	Display which routing protocols are running, their timers, and networks being advertised
router <protocol> <id>	Enter routing protocol configuration mode (e.g., router ospf 1, router eigrp 100)

Command	Purpose
`passive-interface <intf>`	Suppress protocol hellos on an interface while still advertising its network (syntax common across IGPs)

Configuration Walkthrough

Compare administrative distance behavior - static route as a backup to OSPF

```
R1(config)#ip route 0.0.0.0 0.0.0.0 192.168.1.1 250
! AD 250 ensures this floating static is used only if OSPF's
! AD-110 default route disappears from the table
```

Worked Scenario

Q: A network mixes OSPF and a static default route to the internet. Both could theoretically supply a default route - which one wins?

Whichever has the lower Administrative Distance, unless the static route is deliberately configured as a floating static with a higher AD. A normal static route has AD 1, beating OSPF's AD 110, so the static route wins by default unless you intentionally raise its AD above 110 to make it a backup that only activates if the OSPF-learned default route disappears.

Common Mistakes / Gotchas

- Confusing Administrative Distance (compares between different protocols) with Metric (compares paths within the same protocol)
- Assuming all distance-vector protocols are classful - RIPv2 is distance-vector but classless
- Forgetting that IGP vs EGP is about scope (within vs between autonomous systems), not which specific algorithm is used

Exam Tips

★ *Be able to instantly recall the AD table: Connected=0, Static=1, eBGP=20, EIGRP=90, OSPF=110, RIP=120, iBGP=200*

★ *Know which protocols are link-state vs distance-vector vs path-vector - this comparison is tested directly*